

EXPERIENCE

• Sipartech

Toulouse, France

Data Analyst

Feb 2025 – Present

Creation of a position among the GIS team. Goal of facilitating data management for the Pre-sales team.

- **Pre-sales Quotation Generator:** Updated the .xlsm worksheets handling pre-sales quotations according to infrastructure PM's experiences. Refactored and improved VBA macros robustness. Halved the 4000 lines legacy code. Heavily simplified output readability.
- **QGIS Plugins:** Built a .kmz generator to GAFAM clients topological standard. Added Simplex Noise for privacy. Automated DICT/ATU utility response workflows, completely replacing Sogelink and saving the company €40k/year in software licensing.
- **MC80a Home One Tydirium:** Architected an end-to-end multi-layer routing and pricing ecosystem, version controlled and packaged (git / docker) to unify and simplify business processes. Built a REST-like API (FastAPI / PostGIS) backend, a PyQGIS plugin front for Pre-sales team, and endpoint documentation (pydantic) for IT department (Dolibarr). Replaced legacy VBA macros and local pg_routing implementation with a server-side SQL routing and pricing engine (PL/pgSQL). Established an ELT pipeline (dbt + Oracle FDW) for live data sync. Added visualization tools (pg_tileserv).

• Sabbatical

Mont-Saint-Michel, France

Spiritual Retreat

Mar 2023 – Dec 2024

As I understood the importance of spiritual strength and sturdiness, I focused on deepening my life experience.

- **Personal Introspection:** Dove into family history, gathered testimonies and knowledge on my cultural heritage.
- **Spiritual Journey:** Begun spiritual education within the Community of Saint Martin in Mont-Saint-Michel.

• SNCF

Lyon, France

Data Integration Intern

Apr 2022 – Sept 2022

TL;DR: Built core notification services and high-volume data ingestion pipelines

- **Notifications:** Service for sending email, push and in-app notifications. Involved in features such as delivery time optimization, tracking, queuing and A/B testing. Built an internal app to run batch campaigns for marketing, etc.
- **Nostos:** Bulk data processing and injection service from Hadoop to Cassandra and provides a thin REST layer on top for serving offline computed data online.
- **Workflows:** Dataduct an open source.

• Lucena Research

Atlanta, GA

Data Scientist

Summer 2012 and 2013

TL;DR: Built core notification services and high-volume data ingestion pipelines

- **Portfolio Management:** Created models for portfolio hedging, portfolio optimization and price forecasting. Also created a strategy backtesting engine used for simulating and backtesting strategies.
- **QuantDesk:** Python backend for a web application used by hedge fund managers for portfolio management.

• Georgia Institute of Technology

Atlanta, GA

Research and Teaching Assistant

Jan 2012 – Dec 2013

TL;DR: Built core notification services and high-volume data ingestion pipelines

- **Research Assistant – Machine Learning:** Research on machine learning for portfolio hedging and replication algorithms. Modeling low-risk & continuous-return strategies. Developed the Python library QSTK.
- **Teaching Assistant – Computational Investing:** The online course on Coursera, had more than 100,000 students enrolled. It was featured on the 11 Alive News and the Atlanta Journal Constitution. Involved in creating assignments, exams and conducting recitation sessions. Also taught the on-campus version of the course.

EDUCATION

- **University of Caen Normandy** Normandy, France
Master of Science in Applied Mathematics, Statistics and Decision Analysis 2019 – 2022
TL;DR: Built core notification services and high-volume data ingestion pipelines
- **University of Caen Normandy** Normandy, France
Bachelor of Science in General Mathematics 2016 – 2019
TL;DR: Built core notification services and high-volume data ingestion pipelines

PROJECTS

- **QuantSoftware Toolkit:** Open source Python library for financial data analysis and machine learning for finance.
- **GitHub Visualization:** Data visualization of Git log data using D3 to analyze project trends over time.
- **Recommendation System:** Music and movie recommender systems using collaborative filtering on public datasets.
- **Mac Setup:** Book that gives step-by-step instructions on setting up developer environment on macOS.